OpsArena: Production Cloud-Native Incident Response RL Environment

TECHNICAL DESIGN DOCUMENT & VERIFIER SPECIFICATION | TEAM_087

Team Code: team_087	**Track:** Software & DevOps / Business Ops	**Evaluated Model:** gpt-6-luna (penpencil.guru)	**Pre-Flight Baselines: 100% Passed (8/8 Tasks in 2.53s)**

1. Executive Summary & Problem Formulation

Modern cloud infrastructure operations demand that autonomous AI agents diagnose, triage, and remediate cascading incidents across distributed microservice topologies under tight Service Level Objectives (SLOs: error rate < 0.1%, p99 latency < 100ms) and operational budgets ($150.00). Existing RL benchmarks either test synthetic toy math problems or supply unconstrained Docker sandboxes vulnerable to prompt injection, sandbox escapes, and reward tampering. **OpsArena** bridges this gap with a deterministic, hardened, discrete-event simulation of **CloudMesh** (8 microservices), a dual-container isolated evaluation harness, an 8-task procedurally seeded incident benchmark, and a mathematically gated continuous multi-factor verifier.

2. Dual-Container Architecture & Sandboxing Threat Model

OpsArena enforces strict process and network isolation between the model-facing harness and the simulated execution sandbox:

• **Network Containment:** The harness container connects to the external model API (internet) and an internal bridge (isolated). The sandbox container runs exclusively on the isolated network (internal: true) with **zero public internet egress**.

• **Zero Host Socket Access:** The Docker daemon socket (/var/run/docker.sock) is strictly never mounted. Process breakout and sibling container manipulation are eliminated by design.

• **Least Privilege:** The sandbox runs as an unprivileged user (UID 1000: sandbox), enforces cap_drop: ALL, no-new-privileges, a 512MB RAM ceiling, and a 256 PIDs limit.

• **Verifier Inaccessibility:** All evaluation criteria, ground truth thresholds, and reward calculation logic reside solely inside the harness. The sandbox exposes only /health, /reset, /tool, and /state, preventing reward tampering.

• **Adversarial Input Sanitization:** All operational tools enforce key-specific schemas. Passing invalid types (e.g. string 'banana' for db_pool_size) returns clean tool errors rather than crashing the verifier. In-flight logs regex-scrub database passwords and JWT tokens with [REDACTED], and outputs are clipped at 4,000 chars.

3. Operational Action Space (8 Schema-Validated Tools)

Tool	Cost	Parameters & Type Bounds	Operational Function & Safeguards
get_topology	$1.00	None	Inspects 8 services: pod replicas, versions, dependencies, and cumulative spend.
query_metrics	$2.00	service, metric (latency, errors, cpu, mem, qps, rps)	Unified metrics engine: returns 5-step timeseries with status (HEALTHY / CRITICAL).
query_logs	$3.00	service, filter (optional regex)	Structured log query with automated secret sanitization and 4,000 char bounds.
scale_service	$2/pod+$3	service, replicas (int: 1 to 20)	Scales pod pool within platform boundaries; rejects non-ints and out-of-range values.
rollback_release	$20.00	service, target_version (string)	Rolls back faulty deployments to verified stable container images in release registry.
update_config	$5.00	service, key, value (key-specific typed)	Key-specific schema: enforces int bounds (pool: 1-100, lock: 0-60k) and booleans.
reroute_traffic	$10.00	service, weight (int: 0 to 100)	Shifts ingress/canary traffic weights (0% to 100%) to safely isolate failing routes.
finish	$0.00	None	Signals completion of incident mitigation and requests formal verification scoring.

4. Multi-Factor Programmatic Reward Formulation

OpsArena computes a multi-factor continuous reward gated by an absolute incident resolution indicator: **Reward = P × [0.45 × S_SLO + 0.25 × R_reg + 0.15 × B_bud + 0.15 × E_eff]**.

• **Progress Multiplier (P ∈ [0.0, 1.0]):** P = 0.00 for no-op/idle agents, guaranteeing a score of **0.0000** ≤ **0.05**. Bounded partial credit (P ∈ (0, 0.5]) is awarded for incremental progress (e.g. partial scaling or pool expansion), and P = 1.00 upon full incident remediation.

• **SLO Recovery (S_SLO ∈ [0, 1]):** Continuous score evaluating post-incident error rate (≤ 0.1%) and p99 latency (≤ 100ms).

• **Platform Stability (R_reg ∈ [0, 1]):** Compares all non-target microservices against their pre-incident baseline health. If an agent downscales, rolls back, or drains an unaffected service, R_reg drops sharply, penalizing regression.

• **Budget Preservation (B_bud ∈ [0, 1]):** Full marks for spend ≤ $50.00, decaying linearly to 0 at the $150.00 operating budget ceiling.

• **Action Efficiency (E_eff ∈ [0, 1]):** Full marks for surgical resolution within ≤ 8 tool turns, decaying linearly up to 40 turns.

5. Procedurally Seeded 8-Task Incident Benchmark Suite

The benchmark features 8 deterministic incident families with procedural parameter variation by seed:

Task ID	Scenario Title	Diff	Simulated Root Cause & Seeded Variations	Target SRE Action	Oracle	No-Op
task_01	Redis Eviction Storm	Easy	Mem capacity (1-4GB), surge QPS (2.8k-3.6k) causes DB CPU 96%.	Scale cache-cluster ≥ 2 pods	**0.996**	**0.000**
task_02	Bad Deployment Rollback	Easy	Faulty release tag throws NullPointerExceptions (16-21% 500s).	Rollback order-service to v2.4.0	**0.996**	**0.000**
task_03	DB Pool Starvation	Med	Checkout bursts (1.6k-2.2k RPS) exhaust pool (504 timeouts).	Update db_pool_size ≥ 25	**0.996**	**0.000**
task_04	Slow Third-Party Gateway	Med	External payment partner hangs (6s-9s), causing thread starvation.	Enable circuit_breaker_enabled	**0.996**	**0.000**
task_05	Cascading Failure	Hard	Flash surge (4.4k-5.2k RPS) overwhelms pods, causing OOM crash loops.	Scale order-service ≥ 6 pods	**0.994**	**0.000**
task_06	Canary Memory Leak	Med	Severe memory leak on canary split route (5-15% traffic).	Reroute traffic weight to 0%	**0.996**	**0.000**
task_07	Overprovisioned Drain	Hard	Cluster overscaled (36 pods), burning $540/hr under 12% CPU.	Right-size cluster to ≤ 18 pods	**0.996**	**0.000**
task_08	Migration Lock Starvation	Hard	Exclusive table lock blocks writes (22% checkout errors).	Set lock_timeout_ms to 1000	**0.996**	**0.000**

6. Empirical Evaluation & Live Autonomous Model Runs

Pre-Flight Baseline Gating: Across all 8 tasks, pre-flight checks completed in **2.53 seconds** (budget: 120s), with **100% pass rate** (Oracle Mean: **0.996** ≥ **0.95**, No-Op Mean: **0.0000** ≤ **0.05**). Determinism verified across multiple seeds (0, 42, 99).
Live gpt-6-luna Autonomous Evaluations (penpencil.guru): Three real model runs were executed and archived in test-run-logs/:
1. **run_01_task_02_bad_deployment:** Diagnosed NPE in v2.4.1, rolled back to v2.4.0, verified error rate dropped from 18.4% → 0.04% in 4 turns (13.7s, Reward: **0.9955**).
2. **run_02_task_01_cache_pressure:** Investigated cache eviction storm, scaled cache-cluster to 2 pods, verified CPU normalized in 4 turns (14.2s, Reward: **0.9955**).
3. **run_03_task_05_cascading_failure (Negative Control):** Reached step cap during investigative telemetry reads before issuing scale command (Reward: **0.0000**).
Sample Trajectories: Documented in sample-trajectories/trajectory_success_task_02.md and sample-trajectories/trajectory_failed_task_05.md with full SRE post-mortems.

7. Turnkey Verification & Official Run Contract Compliance

Official Hackathon Contract	Required Parameter	OpsArena Implementation & Verification Status
Single-Command Execution	docker compose up --build ...	VERIFIED: docker-compose.yml runs sandbox on isolated net and harness on internet.
Pre-Flight Baseline Gating	Oracle ≥ 0.95, No-Op ≤ 0.05	VERIFIED: Exit code 2 if baseline fails; 8/8 tasks verified in gate.json (2.53s).
Three Real Model Runs	test-run-logs/ (>= 3 runs)	VERIFIED: 3 genuine gpt-6-luna runs archived with JSONL and summary reports.
Success & Failure Trajectories	sample-trajectories/ (>= 2)	VERIFIED: 1 success (task_02, 0.9955) + 1 failure (task_05, 0.0000) with post-mortems.
Resource & Step Boundaries	40 steps, 10 min, 10 tools/turn	VERIFIED: 48k token context trimming, 4k char clipping, 60s tool timeout.
API Key Protocol	.env excluded from archive	VERIFIED: .env ignored by .gitignore; excluded from team_087.zip.